

Solución tarea 12

- Abrimos el archivo de hadoop

```
(kali@nodo1)-[~]  
$ nano /home/kali/.bashrc
```

- Al final del archivo añadimos lo siguiente

```
export SQOOP_HOME=/home/kali/Desktop/hadoop/sqoop  
export PATH=$PATH:$SQOOP_HOME/bin
```

- Aplicamos los cambios

```
(kali@nodo1)-[~]  
$ source /home/kali/.bashrc
```

- Para la instalación de Oracle hemos tenido que crear un contenedor de Docker para poder ejecutarlo puesto que estaba dando demasiados errores.

```
bash  
  
sudo docker run -d --name oracle-xe \  
-p 1521:1521 \  
-e ORACLE_PASSWORD=oracle \  
gvenzl/oracle-xe:11
```

- Iniciamos sesión como SYSTEM

```
SQL*Plus: Release 19.0.0.0.0 - Production on Fri Apr 17 05:36:01 2026  
Version 19.6.0.0.0  
Copyright (c) 1982, 2019, Oracle. All rights reserved.  
  
Connected to:  
Oracle Database 11g Express Edition Release 11.2.0.2.0 - 64bit Production  
SQL>
```

- Creamos las tablas de ejemplo que vienen en la práctica
Tabla Departamentos

```
SQL> CREATE TABLE HR.DEPARTMENTS (  
  DEPARTMENT_ID NUMBER(4),  
  DEPARTMENT_NAME VARCHAR2(30),  
  MANAGER_ID NUMBER(6),  
  LOCATION_ID NUMBER(4)  
); 2 3 4 5 6  
  
Table created.
```

Datos de la tabla departamentos

[illegible]

Tabla empleados

```
SQL> CREATE TABLE HR.EMPLOYEES (  
  EMPLOYEE_ID NUMBER(6),  
  FIRST_NAME VARCHAR2(20),  
  LAST_NAME VARCHAR2(25),  
  EMAIL VARCHAR2(25),  
  PHONE_NUMBER VARCHAR2(20),  
  HIRE_DATE DATE,  
  JOB_ID VARCHAR2(10),  
  SALARY NUMBER(8,2),  
  COMMISSION_PCT NUMBER(2,2),  
  MANAGER_ID NUMBER(6),  
  DEPARTMENT_ID NUMBER(4)  
); 2 3 4 5 6 7 8 9 10 11 12 13  
  
Table created.  
  
SQL> █
```

Datos de la tabla empleados

```
SQL> INSERT INTO HR.EMPLOYEES VALUES (100, 'Steven', 'King', 'SKING', '515.123.4567', SYSDATE, 'AD_PRES', 24000, NULL, NULL, 90);  
INSERT INTO HR.EMPLOYEES VALUES (114, 'Den', 'Raphaely', 'DRAPHEAL', '515.127.4561', SYSDATE, 'PU_MAN', 11000, NULL, 100, 30);  
INSERT INTO HR.EMPLOYEES VALUES (121, 'Adam', 'Frippe', 'AFRIPP', '650.123.2234', SYSDATE, 'ST_MAN', 8200, NULL, 100, 50);  
INSERT INTO HR.EMPLOYEES VALUES (145, 'John', 'Russell', 'JRUSSEL', '011.44.1344.429268', SYSDATE, 'SA_MAN', 14000, 0.4, 100, 80);  
INSERT INTO HR.EMPLOYEES VALUES (200, 'Jennifer', 'Whalen', 'JWHALEN', '515.123.4444', SYSDATE, 'AD_ASST', 4400, NULL, 101, 10);  
INSERT INTO HR.EMPLOYEES VALUES (201, 'Michael', 'Hartstein', 'MHARTSTE', '515.123.5555', SYSDATE, 'MK_MAN', 13000, NULL, 100, 20);  
INSERT INTO HR.EMPLOYEES VALUES (203, 'Susan', 'Mavris', 'SMAVRIS', '515.123.7777', SYSDATE, 'HR_REP', 6500, NULL, 101, 40);  
INSERT INTO HR.EMPLOYEES VALUES (204, 'Hermann', 'Baer', 'HBAER', '515.123.8888', SYSDATE, 'PR_REP', 10000, NULL, 101, 70);  
INSERT INTO HR.EMPLOYEES VALUES (205, 'Shelley', 'Higgins', 'SHIGGINS', '515.123.8080', SYSDATE, 'AC_MGR', 12000, NULL, 101, 110);  
INSERT INTO HR.EMPLOYEES VALUES (192, 'Alexis', 'Bull', 'ABULL', '650.127.4234', SYSDATE, 'SH_CLERK', 4100, NULL, 121, 50);  
COMMIT;  
1 row created.  
  
SQL>  
1 row created.  
  
SQL>  
1 row created.  
  
SQL>  
1 row created.  
  
SQL>  
1 row created.  
  
SQL>  
1 row created.  
  
SQL>  
1 row created.  
  
SQL>  
1 row created.  
  
SQL>  
1 row created.  
  
SQL>  
1 row created.  
  
SQL> █
```

- Instalamos Sqoop

```
\\[e]@\\[u]h: \\[a]\\.\\[033[1;31m]u@\\[033[94m]--\\[033[0;1m]w\\[033[94m]n\\[033[94m]--\\[033[1;31m]s\\[033[0m] wget https://archive.apache.org/dist/sqoop/1.4.7/sqoop-1.4.7.bin_hadoop-2.6.0.tar.gz  
-2026-04-17 05:14:28- https://archive.apache.org/dist/sqoop/1.4.7/sqoop-1.4.7.bin_hadoop-2.6.0.tar.gz  
Resolving archive.apache.org (archive.apache.org)... 65.180.204.189, 2401:4921:a805::2  
Connecting to archive.apache.org (archive.apache.org)[65.180.204.189]:443... connected.  
HTTP request sent, awaiting response... 200 OK  
Length: 1795360 (1.7M) [application/x-gzip]  
Saving to: 'sqoop-1.4.7.bin_hadoop-2.6.0.tar.gz'  
  
sqoop-1.4.7.bin_hadoop-2.6.0.tar.gz 3K[====] 578.01K 254KB/s █
```

- Lo descomprimos

```
\\[e]@\\[u]h: \\[a]\\.\\[033[1;31m]u@\\[033[94m]--\\[033[0;1m]w\\[033[94m]n\\[033[94m]--\\[033[1;31m]s\\[033[0m] tar -xzf sqoop-1.4.7.bin_hadoop-2.6.0.tar.gz  
\\[e]@\\[u]h: \\[a]\\.\\[033[1;31m]u@\\[033[94m]--\\[033[0;1m]w\\[033[94m]n\\[033[94m]--\\[033[1;31m]s\\[033[0m] █
```

- Lo movemos a la ruta correcta según la práctica

```
\[e]0;u@h: \w[a]\[\033[94m]└─(\[\033[1;31m]u@ \h\[\033[94m])-(\[\033[0;1m]w\[\033[94m])\n\[\033[94m]└─\[\033[1;31m]$\[\033[0m] sudo mv sqoop-1.4.7.bin_hadoop-2.6.0 /opt/hadoop/sqoop
```

- Comprobamos que sqoop funciona

```
\[e]0;u@h: \w[a]\[\033[94m]└─(\[\033[1;31m]u@ \h\[\033[94m])-(\[\033[0;1m]w\[\033[94m])\n\[\033[94m]└─\[\033[1;31m]$\[\033[0m] sqoop version
Warning: /opt/hadoop/sqoop/./hbase does not exist! HBase imports will fail.
Please set $HBASE_HOME to the root of your HBase installation.
Warning: /opt/hadoop/sqoop/./hcatalog does not exist! HCatalog jobs will fail.
Please set $HCAT_HOME to the root of your HCatalog installation.
Warning: /opt/hadoop/sqoop/./accumulo does not exist! Accumulo imports will fail.
Please set $ACCUMULO_HOME to the root of your Accumulo installation.
Warning: /opt/hadoop/sqoop/./zookeeper does not exist! Accumulo imports will fail.
Please set $ZOOKEEPER_HOME to the root of your Zookeeper installation.
SLF4J: Failed to load class "org.slf4j.impl.StaticLoggerBinder".
SLF4J: Defaulting to no-operation (NOP) logger implementation
SLF4J: See http://www.slf4j.org/codes.html#StaticLoggerBinder for further details.
2026-04-17 07:26:57,826 INFO sqoop.Sqoop: Running Sqoop version: 1.4.7
Sqoop 1.4.7
git commit id 2328971411f57f0cb683dfb79d19d4d19d185dd8
Compiled by maugli on Thu Dec 21 15:59:58 STD 2017

\[e]0;u@h: \w[a]\[\033[94m]└─(\[\033[1;31m]u@ \h\[\033[94m])-(\[\033[0;1m]w\[\033[94m])\n\[\033[94m]└─\[\033[1;31m]$\[\033[0m] █
```

- Iniciamos los servicios de Hadoop

```
\[e]0;u@h: \w[a]\[\033[32m]└─(\[\033[1;34m]u@ \h\[\033[32m])-(\[\033[0;1m]w\[\033[32m])\n\[\033[32m]└─\[\033[1;34m]$\[\033[0m] /home/kali/Desktop/hadoop/sbin/start-dfs.sh
Starting namenodes on [nodo1]
Starting datanodes
nodo2: ssh: connect to host nodo2 port 22: No route to host
Starting secondary namenodes [nodo1]
SLF4J: Failed to load class "org.slf4j.impl.StaticLoggerBinder".
SLF4J: Defaulting to no-operation (NOP) logger implementation
SLF4J: See http://www.slf4j.org/codes.html#StaticLoggerBinder for further details.

\[e]0;u@h: \w[a]\[\033[32m]└─(\[\033[1;34m]u@ \h\[\033[32m])-(\[\033[0;1m]w\[\033[32m])\n\[\033[32m]└─\[\033[1;34m]$\[\033[0m] █

\[e]0;u@h: \w[a]\[\033[32m]└─(\[\033[1;34m]u@ \h\[\033[32m])-(\[\033[0;1m]w\[\033[32m])\n\[\033[32m]└─\[\033[1;34m]$\[\033[0m] /home/kali/Desktop/hadoop/sbin/start-yarn.sh
Starting resourcemanager
Starting node-managers
nodo2: ssh: connect to host nodo2 port 22: No route to host

\[e]0;u@h: \w[a]\[\033[32m]└─(\[\033[1;34m]u@ \h\[\033[32m])-(\[\033[0;1m]w\[\033[32m])\n\[\033[32m]└─\[\033[1;34m]$\[\033[0m] █
```

- Entramos con Sqoop a la base de datos de Oracle, leemos la tabla departments y guarda los 11 registros que había en la tabla

```
\[e]0;u@h: \w[a]\[\033[32m]└─(\[\033[1;34m]u@ \h\[\033[32m])-(\[\033[0;1m]w\[\033[32m])\n\[\033[32m]└─\[\033[1;34m]$\[\033[0m] sqoop-import \
--connect jdbc:oracle:thin:@127.0.0.1:1521:XE \
--username HR --password HR \
--table DEPARTMENTS \
--target-dir /empleados \
-m 1
Warning: /opt/hadoop/sqoop/./hbase does not exist! HBase imports will fail.
Please set $HBASE_HOME to the root of your HBase installation.
Warning: /opt/hadoop/sqoop/./hcatalog does not exist! HCatalog jobs will fail.
Please set $HCAT_HOME to the root of your HCatalog installation.
Warning: /opt/hadoop/sqoop/./accumulo does not exist! Accumulo imports will fail.
Please set $ACCUMULO_HOME to the root of your Accumulo installation.
Warning: /opt/hadoop/sqoop/./zookeeper does not exist! Accumulo imports will fail.
Please set $ZOOKEEPER_HOME to the root of your Zookeeper installation.
SLF4J: Failed to load class "org.slf4j.impl.StaticLoggerBinder".
SLF4J: Defaulting to no-operation (NOP) logger implementation
SLF4J: See http://www.slf4j.org/codes.html#StaticLoggerBinder for further details.
2026-04-19 10:50:12,110 INFO sqoop.Sqoop: Running Sqoop version: 1.4.7
2026-04-19 10:50:12,175 WARN tool.BaseSqoopTool: Setting your password on the command-line is insecure. Consider using -P instead.
2026-04-19 10:50:12,294 INFO oracle.OraOopManagerFactory: Data Connector for Oracle and Hadoop is disabled.
2026-04-19 10:50:12,312 INFO manager.SqlManager: Using default fetchSize of 1000
2026-04-19 10:50:12,313 INFO tool.CodeGenTool: Beginning code generation
2026-04-19 10:50:12,831 INFO manager.OracleManager: Time zone has been set to GMT
2026-04-19 10:50:12,875 INFO manager.SqlManager: Executing SQL statement: SELECT t.* FROM DEPARTMENTS t WHERE 1=0
2026-04-19 10:50:13,039 INFO orm.CompilationManager: HADOOP_MAPRED_HOME is /home/kali/Desktop/hadoop
Note: /tmp/sqoop-kali/compile/0f6b576a84217a333f652be39c3a3d0c/DEPARTMENTS.java uses or overrides a deprecated API.
Note: Recompile with -Xlint:deprecation for details.
2026-04-19 10:50:14,850 INFO orm.CompilationManager: Writing jar file: /tmp/sqoop-kali/compile/0f6b576a84217a333f652be39c3a3d0c/DEPARTMENTS.jar
2026-04-19 10:50:14,876 INFO manager.OracleManager: Time zone has been set to GMT
2026-04-19 10:50:14,889 INFO manager.OracleManager: Time zone has been set to GMT
2026-04-19 10:50:14,894 INFO mapreduce.ImportJobBase: Beginning import of DEPARTMENTS
2026-04-19 10:50:15,074 INFO manager.OracleManager: Time zone has been set to GMT
2026-04-19 10:50:27,404 INFO db.DBInputFormat: Using read committed transaction isolation
2026-04-19 10:50:51,665 INFO mapreduce.ImportJobBase: Transferred 250 bytes in 36.0254 seconds (6.9396 bytes/sec)
2026-04-19 10:50:51,671 INFO mapreduce.ImportJobBase: Retrieved 11 records.

\[e]0;u@h: \w[a]\[\033[32m]└─(\[\033[1;34m]u@ \h\[\033[32m])-(\[\033[0;1m]w\[\033[32m])\n\[\033[32m]└─\[\033[1;34m]$\[\033[0m] s█
```

- Verificamos que los ficheros están en Hadoop

```
\[e]0;\u0h: \w[a]\[\033[;32m\]└─(\[\033[1;34m\]\u0 \h\[\033[;32m\])-(\[\033[0;1m\]\w\[\033[;32m\])\n\[\033[;32m\]└─\[\033[1;34m\]$\[\033[0m\] sqoop-import \
--connect jdbc:oracle:thin:@127.0.0.1:1521:XE \
--username HR --password HR \
--table DEPARTMENTS \
--target-dir /empleados \
-m 1
Warning: /opt/hadoop/sqoop/../hbase does not exist! HBase imports will fail.
Please set $HBASE_HOME to the root of your HBase installation.
Warning: /opt/hadoop/sqoop/../hcatalog does not exist! HCatalog jobs will fail.
Please set $HCAT_HOME to the root of your HCatalog installation.
Warning: /opt/hadoop/sqoop/../accumulo does not exist! Accumulo imports will fail.
Please set $ACCUMULO_HOME to the root of your Accumulo installation.
Warning: /opt/hadoop/sqoop/../zookeeper does not exist! Accumulo imports will fail.
Please set $ZOOKEEPER_HOME to the root of your Zookeeper installation.
SLF4J: Failed to load class "org.slf4j.impl.StaticLoggerBinder".
SLF4J: Defaulting to no-operation (NOP) logger implementation
SLF4J: See http://www.slf4j.org/codes.html#StaticLoggerBinder for further details.
2026-04-19 10:50:12,110 INFO sqoop.Sqoop: Running Sqoop version: 1.4.7
2026-04-19 10:50:12,175 WARN tool.BaseSqoopTool: Setting your password on the command-line is insecure. Consider using -P instead.
2026-04-19 10:50:12,294 INFO oracle.OraOopManagerFactory: Data Connector for Oracle and Hadoop is disabled.
2026-04-19 10:50:12,312 INFO manager.SqlManager: Using default fetchSize of 1000
2026-04-19 10:50:12,313 INFO tool.CodeGenTool: Beginning code generation
2026-04-19 10:50:12,831 INFO manager.OracleManager: Time zone has been set to GMT
2026-04-19 10:50:12,875 INFO manager.SqlManager: Executing SQL statement: SELECT t.* FROM DEPARTMENTS t WHERE 1=0
2026-04-19 10:50:13,039 INFO orm.CompilationManager: HADOOP_MAPRED_HOME is /home/kali/Desktop/hadoop
Note: /tmp/sqoop-kali/compile/0f6b576a84217a333f652be39c3a3d0c/DEPARTMENTS.java uses or overrides a deprecated API.
Note: Recompile with -Xlint:deprecation for details.
2026-04-19 10:50:14,850 INFO orm.CompilationManager: Writing jar file: /tmp/sqoop-kali/compile/0f6b576a84217a333f652be39c3a3d0c/DEPARTMENTS.jar
2026-04-19 10:50:14,876 INFO manager.OracleManager: Time zone has been set to GMT
2026-04-19 10:50:14,889 INFO manager.OracleManager: Time zone has been set to GMT
2026-04-19 10:50:15,074 INFO manager.OracleManager: Beginning import of DEPARTMENTS
2026-04-19 10:50:15,074 INFO manager.OracleManager: Time zone has been set to GMT
2026-04-19 10:50:27,404 INFO db.DBInputFormat: Using read committed transaction isolation
2026-04-19 10:50:51,665 INFO mapreduce.ImportJobBase: Transferred 250 bytes in 36.0254 seconds (6.9396 bytes/sec)
2026-04-19 10:50:51,671 INFO mapreduce.ImportJobBase: Retrieved 11 records.

\[e]0;\u0h: \w[a]\[\033[;32m\]└─(\[\033[1;34m\]\u0 \h\[\033[;32m\])-(\[\033[0;1m\]\w\[\033[;32m\])\n\[\033[;32m\]└─\[\033[1;34m\]$\[\033[0m\] ss
```

- Extraemos los ID del departamento y su nombre, cambiando la carpeta de hadoop a /practicas/departamentos

```
\[e]0;\u0h: \w[a]\[\033[;32m\]└─(\[\033[1;34m\]\u0 \h\[\033[;32m\])-(\[\033[0;1m\]\w\[\033[;32m\])\n\[\033[;32m\]└─\[\033[1;34m\]$\[\033[0m\] sqoop-import \
--connect jdbc:oracle:thin:@127.0.0.1:1521:XE \
--username HR --password HR \
--table DEPARTMENTS \
--columns "DEPARTMENT_ID,DEPARTMENT_NAME" \
--target-dir /practicas/departamentos \
--as-textfile \
-m 1
Warning: /opt/hadoop/sqoop/../hbase does not exist! HBase imports will fail.
Please set $HBASE_HOME to the root of your HBase installation.
Warning: /opt/hadoop/sqoop/../hcatalog does not exist! HCatalog jobs will fail.
Please set $HCAT_HOME to the root of your HCatalog installation.
Warning: /opt/hadoop/sqoop/../accumulo does not exist! Accumulo imports will fail.
Please set $ACCUMULO_HOME to the root of your Accumulo installation.
Warning: /opt/hadoop/sqoop/../zookeeper does not exist! Accumulo imports will fail.
Please set $ZOOKEEPER_HOME to the root of your Zookeeper installation.
SLF4J: Failed to load class "org.slf4j.impl.StaticLoggerBinder".
SLF4J: Defaulting to no-operation (NOP) logger implementation
SLF4J: See http://www.slf4j.org/codes.html#StaticLoggerBinder for further details.
2026-04-19 11:32:30,344 INFO sqoop.Sqoop: Running Sqoop version: 1.4.7
2026-04-19 11:32:30,408 WARN tool.BaseSqoopTool: Setting your password on the command-line is insecure. Consider using -P instead.
2026-04-19 11:32:30,540 INFO oracle.OraOopManagerFactory: Data Connector for Oracle and Hadoop is disabled.
2026-04-19 11:32:30,552 INFO manager.SqlManager: Using default fetchSize of 1000
2026-04-19 11:32:30,553 INFO tool.CodeGenTool: Beginning code generation
2026-04-19 11:32:31,047 INFO manager.OracleManager: Time zone has been set to GMT
2026-04-19 11:32:31,089 INFO manager.SqlManager: Executing SQL statement: SELECT t.* FROM DEPARTMENTS t WHERE 1=0
2026-04-19 11:32:31,132 INFO orm.CompilationManager: HADOOP_MAPRED_HOME is /home/kali/Desktop/hadoop
Note: /tmp/sqoop-kali/compile/8b11cee2a3d793c45112cce1332d7d26/DEPARTMENTS.java uses or overrides a deprecated API.
Note: Recompile with -Xlint:deprecation for details.
2026-04-19 11:32:32,792 INFO orm.CompilationManager: Writing jar file: /tmp/sqoop-kali/compile/8b11cee2a3d793c45112cce1332d7d26/DEPARTMENTS.jar
2026-04-19 11:32:32,814 INFO manager.OracleManager: Time zone has been set to GMT
2026-04-19 11:32:32,816 INFO mapreduce.ImportJobBase: Beginning import of DEPARTMENTS
2026-04-19 11:32:48,751 INFO db.DBInputFormat: Using read committed transaction isolation
2026-04-19 11:33:11,673 INFO mapreduce.ImportJobBase: Transferred 151 bytes in 38.1529 seconds (3.9578 bytes/sec)
2026-04-19 11:33:11,685 INFO mapreduce.ImportJobBase: Retrieved 11 records.

\[e]0;\u0h: \w[a]\[\033[;32m\]└─(\[\033[1;34m\]\u0 \h\[\033[;32m\])-(\[\033[0;1m\]\w\[\033[;32m\])\n\[\033[;32m\]└─\[\033[1;34m\]$\[\033[0m\]
```

- Verificamos los resultados

```
\[e]0;\u0h: \w[a]\[\033[;32m\]└─(\[\033[1;34m\]\u0 \h\[\033[;32m\])-(\[\033[0;1m\]\w\[\033[;32m\])\n\[\033[;32m\]└─\[\033[1;34m\]$\[\033[0m\] /home/kali/Desktop/hadoop/bin/hdfs ls /practicas/departamentos
SLF4J: Failed to load class "org.slf4j.impl.StaticLoggerBinder".
SLF4J: Defaulting to no-operation (NOP) logger implementation
SLF4J: See http://www.slf4j.org/codes.html#StaticLoggerBinder for further details.
Found 2 items
-rw-r--r-- 2 kali supergroup 0 2026-04-19 11:33 /practicas/departamentos/SUCCESS
-rw-r--r-- 2 kali supergroup 151 2026-04-19 11:33 /practicas/departamentos/part-m-00000

\[e]0;\u0h: \w[a]\[\033[;32m\]└─(\[\033[1;34m\]\u0 \h\[\033[;32m\])-(\[\033[0;1m\]\w\[\033[;32m\])\n\[\033[;32m\]└─\[\033[1;34m\]$\[\033[0m\] /home/kali/Desktop/hadoop/bin/hdfs cat /practicas/departamentos/part-m-00000
SLF4J: Failed to load class "org.slf4j.impl.StaticLoggerBinder".
SLF4J: Defaulting to no-operation (NOP) logger implementation
SLF4J: See http://www.slf4j.org/codes.html#StaticLoggerBinder for further details.
20.Administration
20.Marketing
30.Purchasing
40.Human Resources
50.Shipping
60.IT
70.Public Relations
80.Sales
90.Executive
100.Finance
110.Accounting
```

- Extraemos los departamentos cuyo id sea mayor a 200

```
\[e]0;u@h: \w[a]\[\033[;32m]─(\[\033[1;34m]\u@ \h\[\033[;32m])-(\[\033[0;1m]\w\[\033[;32m])\n\[\033[;32m]└─\[\033[1;34m]$\[\033[0m] sqoop-import \
--connect jdbc:oracle:thin:@127.0.0.1:1521:XE \
--username HR --password HR \
--table DEPARTMENTS \
--columns "DEPARTMENT_ID,DEPARTMENT_NAME" \
--where "department_id > 200" \
--target-dir /practicas/dept_200 \
--as-textfile \
-m 1
Warning: /opt/hadoop/sqoop../hbase does not exist! HBase imports will fail.
Please set $HBASE_HOME to the root of your HBase installation.
Warning: /opt/hadoop/sqoop../hcatalog does not exist! HCatalog jobs will fail.
Please set $HCAT_HOME to the root of your HCatalog installation.
Warning: /opt/hadoop/sqoop../accumulo does not exist! Accumulo imports will fail.
Please set $ACCUMULO_HOME to the root of your Accumulo installation.
Warning: /opt/hadoop/sqoop../zookeeper does not exist! Accumulo imports will fail.
Please set $ZOOKEEPER_HOME to the root of your Zookeeper installation.
SLF4J: failed to load class "org.slf4j.impl.StaticLoggerBinder".
SLF4J: Defaulting to no-operation (NOP) logger implementation
SLF4J: See http://www.slf4j.org/codes.html#StaticLoggerBinder for further details.
2026-04-19 11:35:16,903 INFO sqoop.Sqoop: Running Sqoop version: 1.4.7
2026-04-19 11:35:16,980 WARN tool.BaseSqoopTool: Setting your password on the command-line is insecure. Consider using -P instead.
2026-04-19 11:35:17,114 INFO oracle.OraOopManagerFactory: Data Connector for Oracle and Hadoop is disabled.
2026-04-19 11:35:17,132 INFO manager.SqlManager: Using default fetchSize of 1000
2026-04-19 11:35:17,132 INFO tool.CodeGenTool: Beginning code generation
2026-04-19 11:35:17,628 INFO manager.OracleManager: Time zone has been set to GMT
2026-04-19 11:35:17,673 INFO manager.SqlManager: Executing SQL statement: SELECT t.* FROM DEPARTMENTS t WHERE 1=0
2026-04-19 11:35:17,711 INFO orm.CompilationManager: HADOOP_MAPRED_HOME is /home/kali/Desktop/hadoop
Note: /tmp/sqoop-kali/compile/350e1af58ab81f623486521611afe8c7/DEPARTMENTS.java uses or overrides a deprecated API.
Note: Recompile with -Xlint:deprecation for details.
2026-04-19 11:35:19,066 INFO orm.CompilationManager: Writing jar file: /tmp/sqoop-kali/compile/350e1af58ab81f623486521611afe8c7/DEPARTMENTS.jar
2026-04-19 11:35:19,994 INFO manager.OracleManager: Time zone has been set to GMT
2026-04-19 11:35:20,000 INFO mapreduce.ImportJobBase: Beginning import of DEPARTMENTS
2026-04-19 11:35:32,717 INFO db.DBInputFormat: Using read committed transaction isolation
2026-04-19 11:36:02,060 INFO mapreduce.ImportJobBase: Transferred 0 bytes in 41.3845 seconds (0 bytes/sec)
2026-04-19 11:36:02,079 INFO mapreduce.ImportJobBase: Retrieved 0 records.

\[e]0;u@h: \w[a]\[\033[;32m]─(\[\033[1;34m]\u@ \h\[\033[;32m])-(\[\033[0;1m]\w\[\033[;32m])\n\[\033[;32m]└─\[\033[1;34m]$\[\033[0m] █
```

- Verificamos la solución. En este caso no hay departamentos con id > 200

```
\[e]0;u@h: \w[a]\[\033[;32m]─(\[\033[1;34m]\u@ \h\[\033[;32m])-(\[\033[0;1m]\w\[\033[;32m])\n\[\033[;32m]└─\[\033[1;34m]$\[\033[0m] /home/kali/Desktop/hadoop/bin/hdfs dfs -ls /practicas/dept_200
SLF4J: failed to load class "org.slf4j.impl.StaticLoggerBinder".
SLF4J: Defaulting to no-operation (NOP) logger implementation
SLF4J: See http://www.slf4j.org/codes.html#StaticLoggerBinder for further details.
Found 2 items
-rw-r--r-- 2 kali supergroup 0 2026-04-19 11:35 /practicas/dept_200/SUCCESS
-rw-r--r-- 2 kali supergroup 0 2026-04-19 11:35 /practicas/dept_200/part-m-00000

\[e]0;u@h: \w[a]\[\033[;32m]─(\[\033[1;34m]\u@ \h\[\033[;32m])-(\[\033[0;1m]\w\[\033[;32m])\n\[\033[;32m]└─\[\033[1;34m]$\[\033[0m] /home/kali/Desktop/hadoop/bin/hdfs dfs -cat /practicas/dept_200/part-m-00000
SLF4J: failed to load class "org.slf4j.impl.StaticLoggerBinder".
SLF4J: Defaulting to no-operation (NOP) logger implementation
SLF4J: See http://www.slf4j.org/codes.html#StaticLoggerBinder for further details.

\[e]0;u@h: \w[a]\[\033[;32m]─(\[\033[1;34m]\u@ \h\[\033[;32m])-(\[\033[0;1m]\w\[\033[;32m])\n\[\033[;32m]└─\[\033[1;34m]$\[\033[0m] █
```

- Mezclamos las dos tablas con una consulta de Join

```
\[e]0;u@h: \w[a]\[\033[;32m]─(\[\033[1;34m]\u@ \h\[\033[;32m])-(\[\033[0;1m]\w\[\033[;32m])\n\[\033[;32m]└─\[\033[1;34m]$\[\033[0m] sqoop-import \
--connect jdbc:oracle:thin:@127.0.0.1:1521:XE \
--username HR --password HR \
--table DEPARTMENTS \
--columns "DEPARTMENT_ID,DEPARTMENT_NAME" \
--target-dir /practicas/departamentos \
--as-textfile \
-m 1
Warning: /opt/hadoop/sqoop../hbase does not exist! HBase imports will fail.
Please set $HBASE_HOME to the root of your HBase installation.
Warning: /opt/hadoop/sqoop../hcatalog does not exist! HCatalog jobs will fail.
Please set $HCAT_HOME to the root of your HCatalog installation.
Warning: /opt/hadoop/sqoop../accumulo does not exist! Accumulo imports will fail.
Please set $ACCUMULO_HOME to the root of your Accumulo installation.
Warning: /opt/hadoop/sqoop../zookeeper does not exist! Accumulo imports will fail.
Please set $ZOOKEEPER_HOME to the root of your Zookeeper installation.
SLF4J: failed to load class "org.slf4j.impl.StaticLoggerBinder".
SLF4J: Defaulting to no-operation (NOP) logger implementation
SLF4J: See http://www.slf4j.org/codes.html#StaticLoggerBinder for further details.
2026-04-19 11:32:30,344 INFO sqoop.Sqoop: Running Sqoop version: 1.4.7
2026-04-19 11:32:30,408 WARN tool.BaseSqoopTool: Setting your password on the command-line is insecure. Consider using -P instead.
2026-04-19 11:32:30,540 INFO oracle.OraOopManagerFactory: Data Connector for Oracle and Hadoop is disabled.
2026-04-19 11:32:30,552 INFO manager.SqlManager: Using default fetchSize of 1000
2026-04-19 11:32:30,553 INFO tool.CodeGenTool: Beginning code generation
2026-04-19 11:32:31,047 INFO manager.OracleManager: Time zone has been set to GMT
2026-04-19 11:32:31,089 INFO manager.SqlManager: Executing SQL statement: SELECT t.* FROM DEPARTMENTS t WHERE 1=0
2026-04-19 11:32:31,132 INFO orm.CompilationManager: HADOOP_MAPRED_HOME is /home/kali/Desktop/hadoop
Note: /tmp/sqoop-kali/compile/8b11cee2a3d793c45112cce1332d7d26/DEPARTMENTS.java uses or overrides a deprecated API.
Note: Recompile with -Xlint:deprecation for details.
2026-04-19 11:32:32,792 INFO orm.CompilationManager: Writing jar file: /tmp/sqoop-kali/compile/8b11cee2a3d793c45112cce1332d7d26/DEPARTMENTS.jar
2026-04-19 11:32:32,814 INFO manager.OracleManager: Time zone has been set to GMT
2026-04-19 11:32:32,816 INFO mapreduce.ImportJobBase: Beginning import of DEPARTMENTS
2026-04-19 11:32:48,751 INFO db.DBInputFormat: Using read committed transaction isolation
2026-04-19 11:33:11,673 INFO mapreduce.ImportJobBase: Transferred 151 bytes in 38.1529 seconds (3.9578 bytes/sec)
2026-04-19 11:33:11,685 INFO mapreduce.ImportJobBase: Retrieved 11 records.

\[e]0;u@h: \w[a]\[\033[;32m]─(\[\033[1;34m]\u@ \h\[\033[;32m])-(\[\033[0;1m]\w\[\033[;32m])\n\[\033[;32m]└─\[\033[1;34m]$\[\033[0m] █
```

- Verificamos la solución

```
\[e]0;u@h: \w[a]\[\033[;32m]─(\[\033[1;34m]\u@ \h\[\033[;32m])-(\[\033[0;1m]\w\[\033[;32m])\n\[\033[;32m]└─\[\033[1;34m]$\[\033[0m] /home/kali/Desktop/hadoop/bin/hdfs dfs -cat /practicas/dept_emple/part-m-0000
SLF4J: failed to load class "org.slf4j.impl.StaticLoggerBinder".
SLF4J: Defaulting to no-operation (NOP) logger implementation
SLF4J: See http://www.slf4j.org/codes.html#StaticLoggerBinder for further details.
Join, sequentially, Purchasing, 11000
Alexis,Bull,Shipping,4180
Adam,Frip,Shipping,4200
Jennifer,Whalen,Administration,4480
Hermann,Baer,Public Relations,10880
Michael,Hartstein,Marketing,13600
John,Russell,Sales,14000
Susan,Mavris,Human Resources,6500
Steven,King,Executive,24080
Shelley,Higgins,Accounting,12800

\[e]0;u@h: \w[a]\[\033[;32m]─(\[\033[1;34m]\u@ \h\[\033[;32m])-(\[\033[0;1m]\w\[\033[;32m])\n\[\033[;32m]└─\[\033[1;34m]$\[\033[0m] █
```


- Creamos una tabla vacía en Oracle

```
\[ve0;\u00h: \w\A\[\033[32m\]└─(\[\033[1;34m\]\u@ \h\[\033[32m\])-(\[\033[0;1m\]\w\[\033[32m\])\n\[\033[32m\]└─(\[\033[1;34m\]\$[\033[0m\]) /usr/lib/oracle/19.6/client64/bin/sqlplus HR/HR@127.0.0.1:1521/XE

SQL> SQLPlus: Release 19.0.0.0.0 - Production on Sun Apr 19 11:44:22 2026
Version 19.6.0.0.0

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Connected to:
Oracle Database 11g Express Edition Release 11.2.0.2.0 - 64bit Production

SQL> CREATE TABLE DEPARTAMENTOS_EXPORT (DEPARTMENT_ID NUMBER(4), DEPARTMENT_NAME VARCHAR2(30));
exit;
Table created.

SQL>
```

- Ejecutamos Sqoop para que coja los datos que guardamos en /practicass/departamentos y los meta en una tabla nueva

```
\[ve0;\u00h: \w\A\[\033[32m\]└─(\[\033[1;34m\]\u@ \h\[\033[32m\])-(\[\033[0;1m\]\w\[\033[32m\])\n\[\033[32m\]└─(\[\033[1;34m\]\$[\033[0m\]) sqoop export \
--connect jdbc:oracle:thin:@127.0.0.1:1521:XE \
--username HR --password HR \
--table DEPARTAMENTOS_EXPORT \
--export-dir /practicass/departamentos \
--input-fields-terminated-by ',' \
-m 1
Warning: /opt/hadoop/sqoop/.. /hbase does not exist! HBase imports will fail.
Please set $HBASE_HOME to the root of your HBase installation.
Warning: /opt/hadoop/sqoop/.. /hcatalog does not exist! HCatalog jobs will fail.
Please set $HCAT_HOME to the root of your HCatalog installation.
Warning: /opt/hadoop/sqoop/.. /accumulo does not exist! Accumulo imports will fail.
Please set $ACCUMULO_HOME to the root of your Accumulo installation.
Warning: /opt/hadoop/sqoop/.. /zookeeper does not exist! Accumulo imports will fail.
Please set $ZOOKEEPER_HOME to the root of your Zookeeper installation.
SLF4J: Failed to load class "org.slf4j.impl.StaticLoggerBinder".
SLF4J: Defaulting to no-operation (NOP) logger implementation
SLF4J: See http://www.slf4j.org/codes.html#StaticLoggerBinder for further details.
2026-04-19 11:46:10,904 INFO sqoop.Sqoop: Running Sqoop version: 1.4.7
2026-04-19 11:46:10,980 WARN tool.BaseSqoopTool: Setting your password on the command-line is insecure. Consider using -P instead.
2026-04-19 11:46:11,102 INFO oracle.OraOopManagerFactory: Data Connector for Oracle and Hadoop is disabled.
2026-04-19 11:46:11,144 INFO manager.SqlManager: Using default fetchSize of 1000
2026-04-19 11:46:11,147 INFO tool.CodeGenTool: Beginning code generation
2026-04-19 11:46:11,585 INFO manager.OracleManager: Time zone has been set to GMT
2026-04-19 11:46:11,624 INFO manager.SqlManager: Executing SQL statement: SELECT t.* FROM DEPARTAMENTOS_EXPORT t WHERE 1=0
2026-04-19 11:46:11,698 INFO orm.CompilationManager: HADOOP_MAPRED_HOME is /home/kali/Desktop/hadoop
Note: /tmp/sqoop-kali/compile/ea4beadd232ee1b9a1470277544ca598/DEPARTAMENTOS_EXPORT.java uses or overrides a deprecated API.
Note: Recompile with -Xlint:deprecation for details.
2026-04-19 11:46:13,223 INFO orm.CompilationManager: Writing jar file: /tmp/sqoop-kali/compile/ea4beadd232ee1b9a1470277544ca598/DEPARTAMENTOS_EXPORT.jar
2026-04-19 11:46:13,241 INFO mapreduce.ExportJobBase: Beginning export of DEPARTAMENTOS_EXPORT
2026-04-19 11:46:14,342 INFO manager.OracleManager: Time zone has been set to GMT
2026-04-19 11:46:38,382 INFO mapreduce.ExportJobBase: Transferred 289 bytes in 24.0187 seconds (12.0323 bytes/sec)
2026-04-19 11:46:38,384 INFO mapreduce.ExportJobBase: Exported 11 records.

\[ve0;\u00h: \w\A\[\033[32m\]└─(\[\033[1;34m\]\u@ \h\[\033[32m\])-(\[\033[0;1m\]\w\[\033[32m\])\n\[\033[32m\]└─(\[\033[1;34m\]\$[\033[0m\])
```

- Comprobamos la tabla

```
\[ve0;\u00h: \w\A\[\033[32m\]└─(\[\033[1;34m\]\u@ \h\[\033[32m\])-(\[\033[0;1m\]\w\[\033[32m\])\n\[\033[32m\]└─(\[\033[1;34m\]\$[\033[0m\]) /usr/lib/oracle/19.6/client64/bin/sqlplus HR/HR@127.0.0.1:1521/XE

SQL> SQLPlus: Release 19.0.0.0.0 - Production on Sun Apr 19 11:48:20 2026
Version 19.6.0.0.0

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Connected to:
Oracle Database 11g Express Edition Release 11.2.0.2.0 - 64bit Production

SQL> SELECT * FROM DEPARTAMENTOS_EXPORT;

DEPARTMENT_ID DEPARTMENT_NAME
-----
10 Administration
20 Marketing
30 Purchasing
40 Human Resources
50 Shipping
60 IT
70 Public Relations
80 Sales
90 Executive
100 Finance
110 Accounting

11 rows selected.

SQL>
```

- Creamos una tabla con datos cruzados

```
\[ve0;\u00h: \w\A\[\033[32m\]└─(\[\033[1;34m\]\u@ \h\[\033[32m\])-(\[\033[0;1m\]\w\[\033[32m\])\n\[\033[32m\]└─(\[\033[1;34m\]\$[\033[0m\]) /usr/lib/oracle/19.6/client64/bin/sqlplus HR/HR@127.0.0.1:1521/XE

SQL> SQLPlus: Release 19.0.0.0.0 - Production on Sun Apr 19 11:51:59 2026
Version 19.6.0.0.0

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Connected to:
Oracle Database 11g Express Edition Release 11.2.0.2.0 - 64bit Production

SQL> CREATE TABLE DPT_EMP (nombre VARCHAR2(20), apellidos VARCHAR2(20), departamento VARCHAR2(20), salario NUMBER);
exit;
Table created.

SQL>
```

- Realizamos el comando con sqoop

```
\[v]0;u0h: w[a]\[033[;32m]─\[033[;34m]u0 h\[033[;32m]─\[033[0;1m]w\[033[;32m]]h\[033[;32m]─\[033[;34m]$(\[033[0m]) sqoop export --connect jdbc:oracle:thin:@127.0.0.1:1521:XE \
--username HR --password HR \
--table DPT_EMP \
--export-dir /practicas/dept_emp \
--input-fields-terminated-by ',' \
-m 1
Warning: /opt/hadoop/sqoop/../hbase does not exist! Hbase imports will fail.
Please set $HBASE_HOME to the root of your Hbase installation.
Warning: /opt/hadoop/sqoop/../hcatalog does not exist! HCatalog jobs will fail.
Please set $HCAT_HOME to the root of your HCatalog installation.
Warning: /opt/hadoop/sqoop/../accumulo does not exist! Accumulo imports will fail.
Please set $ACCUMULO_HOME to the root of your Accumulo installation.
Warning: /opt/hadoop/sqoop/../zookeeper does not exist! Accumulo imports will fail.
Please set $ZOOKEEPER_HOME to the root of your Zookeeper installation.
SLF4J: Failed to load class "org.slf4j.impl.StaticLoggerBinder".
SLF4J: Defaulting to no-operation (NOP) logger implementation
SLF4J: See http://www.slf4j.org/codes.html#StaticLoggerBinder for further details.
2020-04-19 11:52:46,934 INFO sqoop.Sqoop: Running Sqoop version: 1.4.7
2020-04-19 11:52:46,998 WARN tool.BaseSqoopTool: Setting your password on the command-line is insecure. Consider using -P instead.
2020-04-19 11:52:47,123 INFO oracle.OracleManagerFactory: Data Connector for Oracle and Hadoop is disabled.
2020-04-19 11:52:47,137 INFO manager.SqlManager: Using default fetchSize of 1000
2020-04-19 11:52:47,138 INFO tool.CodeGenTool: Beginning code generation
2020-04-19 11:52:47,677 INFO manager.OracleManager: Time zone has been set to GMT
2020-04-19 11:52:47,761 INFO manager.SqlManager: Executing SQL statement: SELECT t.* FROM DPT_EMP t WHERE 1=0
2020-04-19 11:52:47,887 INFO orm.CompilationManager: HADOOP_MAPRED_HOME is /home/kali/Desktop/hadoop
Note: /tmp/sqoop-kali/compile/6363453ae21f051251ab027c56ff5858/DPT_EMP.java uses or overrides a deprecated API.
Note: Recompile with -Xlint:deprecation for details.
2020-04-19 11:52:49,699 INFO orm.CompilationManager: Writing jar file: /tmp/sqoop-kali/compile/6363453ae21f051251ab027c56ff5858/DPT_EMP.jar
2020-04-19 11:52:49,726 INFO mapreduce.ExportJobBase: Beginning export of DPT_EMP
2020-04-19 11:52:50,932 INFO manager.OracleManager: Time zone has been set to GMT
2020-04-19 11:53:27,277 INFO mapreduce.ExportJobBase: Transferred 655 bytes in 36.311 seconds (18.0386 bytes/sec)
2020-04-19 11:53:27,282 INFO mapreduce.ExportJobBase: Exported 10 records.
\[v]0;u0h: w[a]\[033[;32m]─\[033[;34m]u0 h\[033[;32m]─\[033[0;1m]w\[033[;32m]]h\[033[;32m]─\[033[;34m]$(\[033[0m])
```